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27 January 1988
SUPERSEDING
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10 March 1965

MILITARY SPECIFICATION

STABILIZER, ORGANIC-TIN-SULFUR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of stabilizing agent containing a tin mercaptide, specifically dibutyl tin beta mercapto proprionate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

RR-S-366	Sieves, Test

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6810

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2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

PUBLICATIONS

Code of Federal Regulations

29 CFR 1910.1200(g) Hazard Communication

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3)

3.2 Physical and chemical properties. The physical and chemical properties of the material shall be as specified in table I.

TABLE I. Physical and chemical properties

Property	Minimum	Maximum
Tin content, percent by weight	35.0	36.0
Sulfur content, percent by weight	9.4	10.0
Melting point, degrees Celsius	115	135
Particle size, percent by weight through a Number 40 U.S. Standard Screen	100	—

3.3 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the precautions cited in the Material Safety Data Sheet. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4 Material safety data sheet (MSDS). The contracting activity shall be provided an MSDS at the time of contract award. The MSDS shall be prepared in accordance with the requirements of 29 CFR 1910.1200(g), Hazard Communication, and included with each shipment. OSHA Form 174 MSDS or a more effective contractor developed MSDS format may be used.

3.5 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. This item must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Toxicological information. The contractor shall have the toxicological formulation and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.3, 3.4, and 6.5).

4.2 Classification of Inspections. The inspections of the material shall be classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Sampling.

4.3.1 First article sample. A first article sample of sufficient quantity shall be selected to perform each of the first article tests detailed in 4.4 (see 6.3).

4.3.2 Quality conformance inspection sampling. Sufficient material shall be taken from each lot to perform the tests specified in 4.5.

4.3.3 Inspection lot. An inspection lot of stabilizer shall consist of all material presented for acceptance at one time and produced in a single manufacturing run under homogeneous conditions of manufacture. The minimum lot size shall be as specified in the contract (see 6.2).

4.4 First article inspection. The first article sample shall be subjected to all the quality conformance inspections detailed in 4.5, at a Government activity designated by the contracting activity (see 6.2). Each test shall be performed in triplicate and the average of each set of three determinations shall be compared with the requirements of 3.2.

4.4.1 First article sample acceptance or rejection. When the results of all the tests of section 4 show that the sample meets all the requirements of sections 3, the first article sample shall be accepted. Otherwise, the first article sample shall be rejected.

4.5 Quality conformance inspection. All quality conformance test samples shall be treated as first article samples (see 6.3). The test methods of 4.5 shall be used to determine that the requirements of this specification have been met. Any change in the test methods employed after acceptance shall have prior written approval of the contracting activity.

4.5.1 Visual examination. Visually examine all containers in the lot and their contents to determine compliance with 3.5 and Section 5.

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4.5.2 Tin content test. Weigh between 1.0 and 1.5 grams (g), to the nearest 0.1 milligram (mg), of sample into a tared high-form porcelain crucible. Add 20 drops of fuming sulfuric acid (A.C.S. grade). Heat the crucible and allow the sample to char. Place the crucible in a muffle furnace and heat to a temperature of 1000° ± 25° Celsius (°C) for not less than 30 minutes. Remove the crucible from the furnace and allow it to cool to room temperature. Weigh the crucible and residue to the nearest 0.1 mg. Calculate the tin content as follows:

$$\text{Tin content, percent} = \frac{W_1 \times 78.764}{W_2}$$

Where

W_1 = Weight of stannic oxide residue, g
W_2 = Weight of sample, g

4.5.2.1 Alternate tin content test. The tin content may be determined by an alternate test method approved by the acquiring activity.

4.5.2.2 Rejection. Failure to meet the tin content requirement of 3.2. shall cause rejection of the lot represented by the sample.

4.5.3 Sulfur content test. The following chemicals and reagents shall be used in this test:

Benzoic acid, A.C.S. grade, powder
Sodium peroxide, CP, calorific grade, 95 percent, minimum
Hydrochloric acid, concentrated, A.C.S. grade
Sulfuric acid, A.C.S. grade, 10-percent aqueous solution
Barium chloride, A.C.S. grade, 10-percent aqueous solution
Ammonium hydroxide, A.C.S. grade, reagent
Potasium perchlorate, A.C.S. grade, powder

4.5.3.1 Preparation of charge. Weigh a clean, dry 22-milliliter (mL) Parr bomb (or equivalent) and add 0.5 + 0.05g of potassium perchlorate and 0.3 ± 0.02g of benzoic acid. Add 0.30 to 0.35g of sample. Add a standard measure of sodium peroxide, and mix thoroughly with a glass or nickel stirring rod.

CAUTION

Mixing with sodium peroxide should be performed behind a safety shield while wearing gloves and a face shield.

4.5.3.2 Ignition of charge.

4.5.3.2.1 Flame ignition method. Place the lid on the bomb and lock it. Place 1 to 2 mL of water in the depression on top of the bomb. Ignite the charge with a Meeker burner, setting the burner so that the hottest part of the flame strikes the bottom of the fusion cup. Shut off the burner when bubbles start to form in the water or if the fusion cup is heated to a dull red color before the water shows any evidence of boiling. Allow the bomb to stand for approximately 2 minutes, then cool in water. Place the bomb behind a safety shield and unlock it, removing cap. Place the bomb on its side in a 600-mL beaker containing approximately 100 mL of distilled water. Warm the water until effersvescence starts. Allow the beaker to cool and wait until effervescence stops, then remove the bomb from the water.

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4.5.3.2.2 Electric ignition method. Place the bomb head containing an ignition wire on the bomb cup and tap the cup gently on the bench a few times. Place the bomb cup in the bell body and tighten the screw cap. Place the assembled bomb under the ignition contact arm in the water bath. Place a safety shield in front of the assembly then fire the bomb with the ignition unit. Allow the bomb to cool in the water bath for at least 5 minutes. Dismantle the bomb, remove the head and dip it into a 600-mL beaker containing approximately 100 mL of distilled water, previously heated almost to boiling. Wash with a fine jet of distilled water, previously heated almost to boiling receiving the washings in the beaker. Lay the fusion cup on its side in the same beaker of water and cover immediately with a watch glass. After the melt has dissolved, remove the cup and wash thoroughly with water, receiving the washings in the same beaker. Allow the solution to cool.

4.5.3.3 Precipitation of barium sulfate. Add concentrated hydrochloric acid to the solution until it is slightly acid, using litmus paper as an indicator. Add concentrated ammonium hydroxide until the solution is slightly alkaline. Heat the solution to almost boiling. Filter and wash the filter paper with hot water, receiving the washings in the same beaker. Add concentrated hydrochloric acid to the filtrate until barely acid. Add 30 ml of barium chloride solution, drop by drop. Heat the solution to boiling and allow it to stand overnight. Filter through a tared Selas, or Gooch, crucible. Wash the precipitate with a 10-percent aqueous solution of sulfuric acid, then wash with alcohol. Dry the crucible in an oven at 105 $\pm$ 10°C for 10 to 15 minutes. Place the crucible in a muffle furnace and heat to a temperature of not less than 800°C. Cool the crucible in a dessicator. Calculate the sulfur content as follows:

$$\text{Sulfur content, percent} = \frac{W_1 \times 13.738}{W_2}$$

where

W_1 = Weight of barium sulfate precipitate, g
W_2 = Weight of sample, g

4.5.3.4 Alternate sulfur content test. The sulfur content may be determined by an alternate method approved by the acquiring activity.

4.5.3.5 Rejection. Failure to meet the sulfur content requirement 3.2 shall cause rejection of the lot represented by the sample.

4.5.4 Melting point test. Fill a melting-point capillary tube with sample material to a height of 3 to 5 millimeters (mm). Attach the capillary tube to a thermometer so that the sample is at the mid-point of the bulb of the thermometer. Suspend the thermometer and capillary tube in a 400-mL beaker containing approximately 300 mL of silicone oil so that the level of oil is at the 51-mm mark. Heat the oil while stirring constantly. When the thermometer indicates 105°C, heat the oil at a rate not greater than 1°C per minute. Record the temperature of the sample when liquid is first observed in the capillary tube and when the sample is completely liquid.

4.5.4.1 Rejection. Failure to meet the melting point requirement of 3.2 shall cause rejection of the lot represented by the sample.

4.5.5 Particle size test. Transfer approximately 100 g of sample weighed to the nearest 0.1 g to a Number 40 U.S. Standard Sieve in accordance with RR-S-366. Vibrate the sample for the five minutes. After vibrating, transfer the material retained on the sieve to a tared waxed paper. Weigh the material to the nearest 0.1 g. Calculate the percent of material retained on the sieve as follows:

$$\text{Weight of material, percent} = \frac{W_1 \times 100}{W_2}$$

where

W_1 = Weight retained on sieve, g
W_2 = Weight of sample, g

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4.5.5.1 Rejection. Failure to meet the particle size requirement of 3.2 shall cause rejection of the lot represented by the sample.

5. **PACKAGING.** (The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation shall be levels A, B, or C, as specified (see 6.2).

5.1.1 Preservation, packaging and packing. Unless otherwise specified by the contracting activity (see 6.2), the preservation, packaging, and packing shall be in accordance with MIL-P-116 method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number, and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material is intended to be used as a stabilizer for propellants for rocket motors.

6.2 Ordering data. Acquisition documents should specify, but not be limited to, the following information:

a. Title, number, and date of this specification
b. Place of delivery
c. Whether a first article sample is required (see 3.1 and 6.3)
d. FAR clauses 23.303 and 52.223-3, if required (see 3.4 and 6.5)
e. Minimum lot size (see 4.3.3)
f. Laboratory to perform first article tests (see 4.4)
g. Preservation level desired (see 5.1).
h. Whether preservation, packaging, or packing is required (see 5.1.1).

6.3 First article sample. When a first article sample inspection is required, the material should be a first article sample from the contractor's current inventory. The first article sample should consist of enough material to perform all tests of 4.5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for additional tests, regular examinations, approval of first article sample test results and disposition of the first article sample. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article sample inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term listing.

Analyses
Melting Point
Particle Size

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6.5 Toxicological formulation and material safety data sheet (MSDS). Material safety data sheet requirements are applicable and should be specified in the contract as required by DOD FAR 23.303. Contracting officers will identify those activities requiring copies of MSDS's. Additional Govenment mailing information is contained in Appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract (see 6.2).

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-81246A(OS)	STABILIZER, ORGANIC-TIN-SULFUR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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